Azure VPN Gateway

Connecting networks together

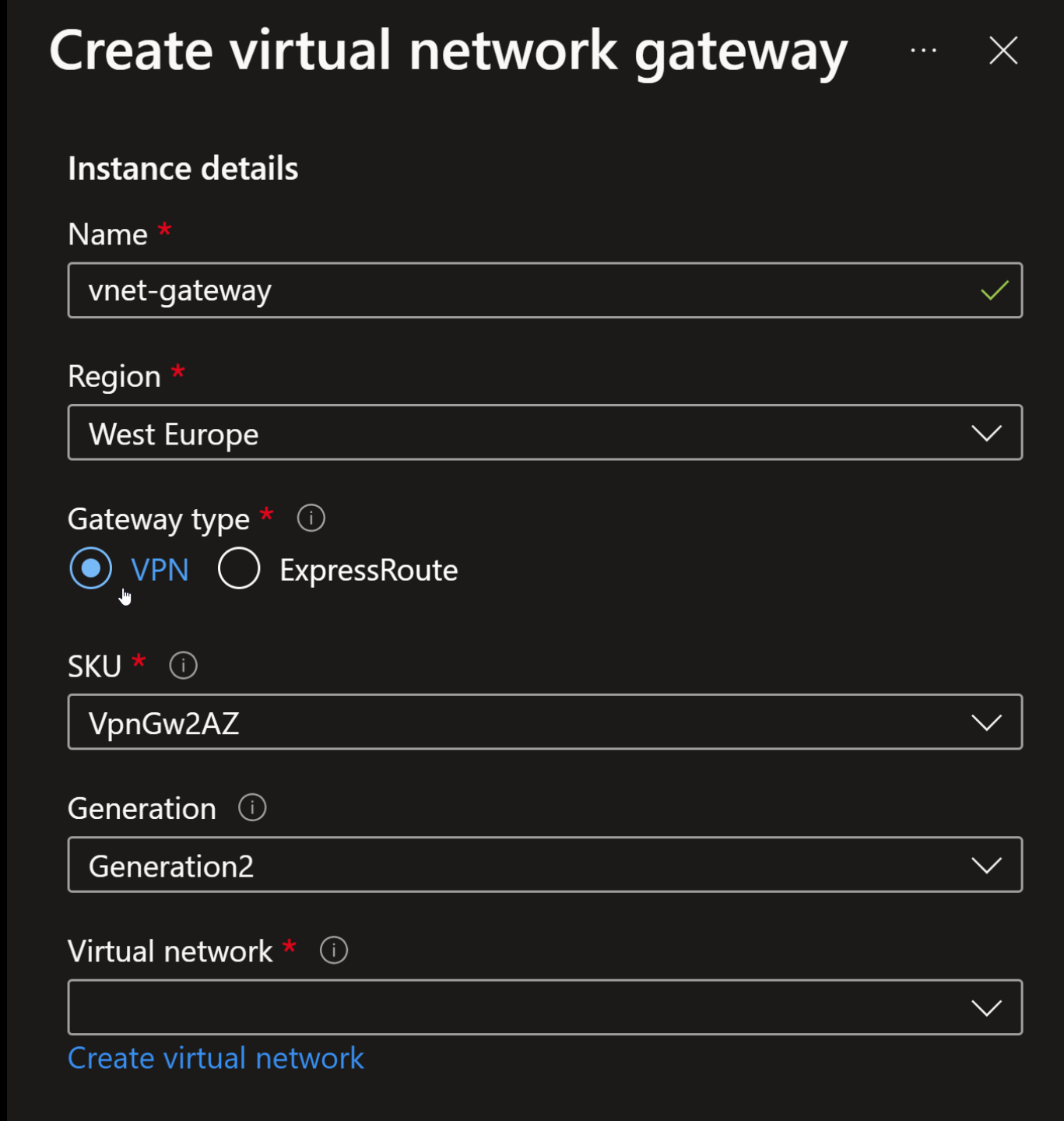
VNET Gateway types

An Azure virtual network gateway could be:

- VPN gateway
- ExpressRoute gateway

An Azure virtual network can have two virtual network gateways:

- one VPN gateway
- one ExpressRoute gateway



Introduction to VPN Gateway

Allow you to set up the following connections:

- Site to Site (S2S)

  Connect on-premises datacenters to Azure virtual networks

- Point-to-site (P2S)

  Connect individual devices to Azure virtual networks

- Network-to-network (prefer peering for cost reasons)

  Connect Azure virtual networks to each other

Site to Site (S2S) VPN Gateway

Connection type *

VNet-to-VNet

VNet-to-VNet

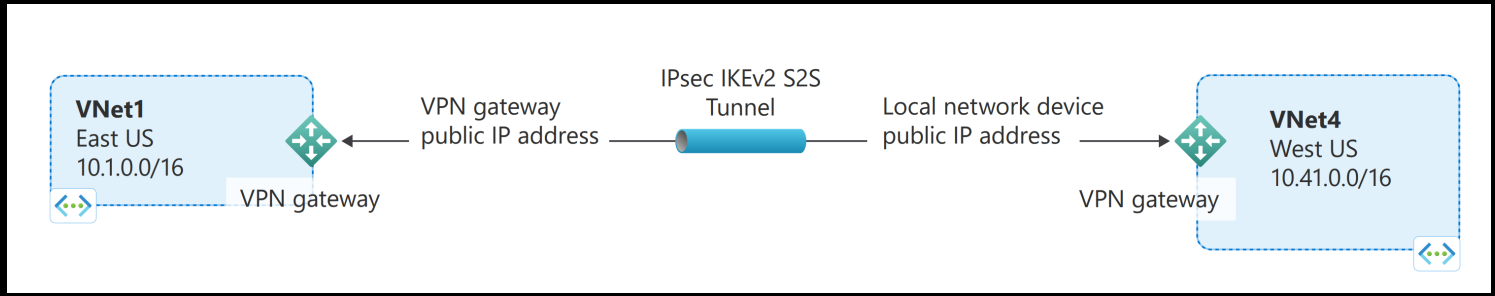
Site-to-site (IPsec)

ExpressRoute

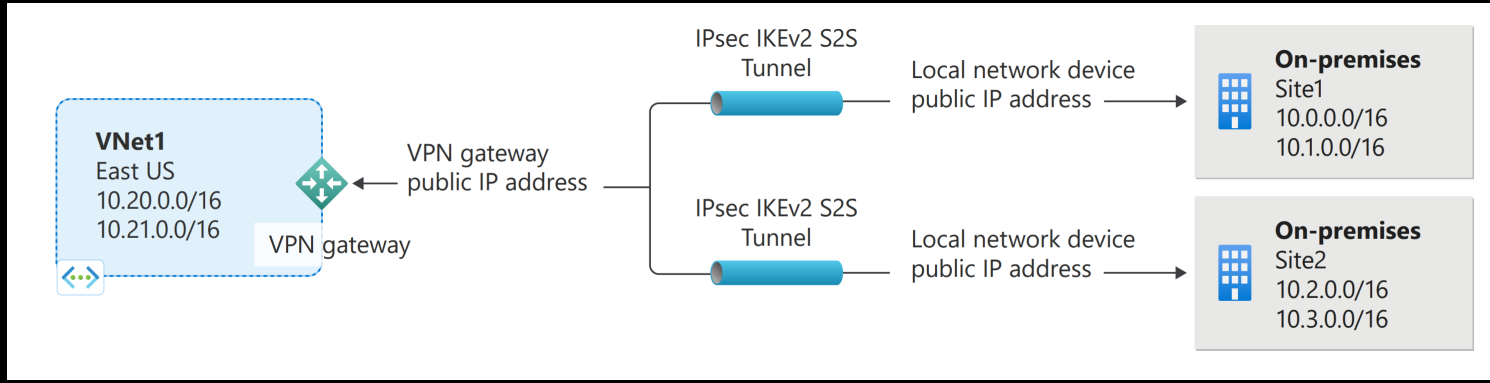
Connect networks with each other through an encrypted VPN tunnel.

Commonly used to:

- Connect on-premises sites to Azure
- Azure VNETs to each other (like VNET Peering)



A S2S connection requires a VPN device located on-premises that has a public IP address assigned to it.



Point to Site (P2S) VPN Gateway

Secure connection to your VNET from an individual client computer.

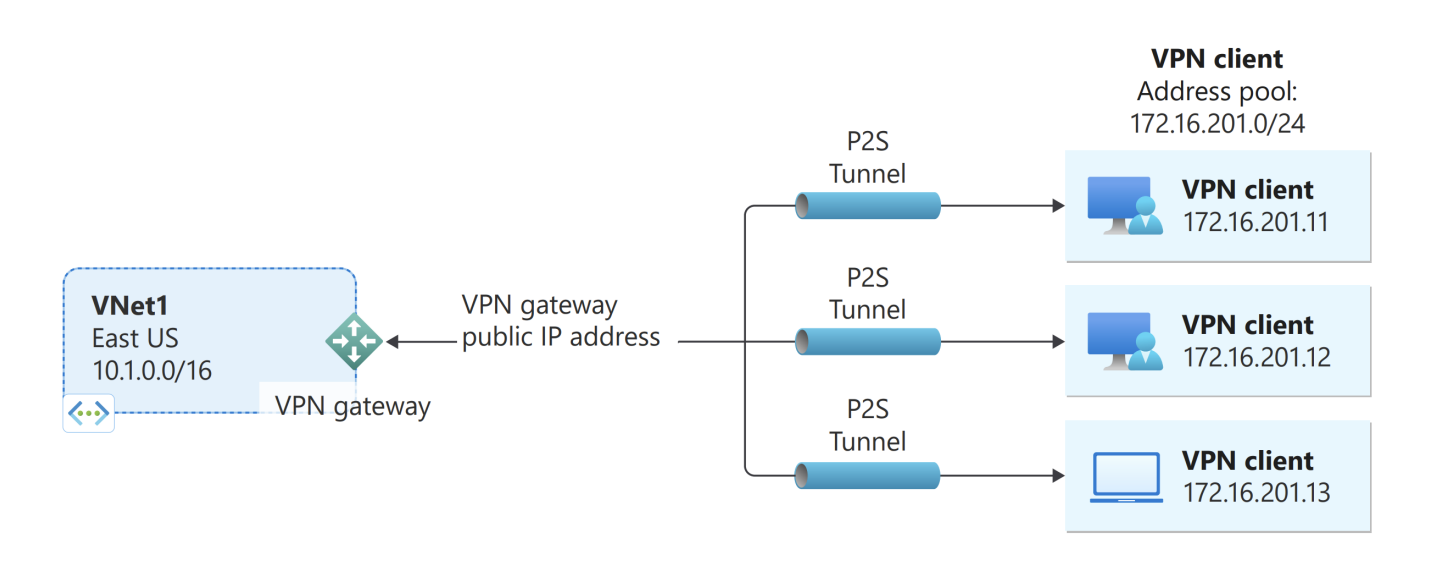
Use instead of S2S VPN when you have only a few clients that need to connect to a VNet.

Supports the following protocols:

- OpenVPN
- SSTP
- IKEv2 VPN

User has to be authenticated with one of

- Certificate authentication
- Microsoft Entra authentication
- Active Directory (AD) Domain Server



VPN Gateway resource

Composed of two or more special VMs deployed to a specific subnet called the GatewaySubnet.

These VMs host routing tables and run specific gateway services.

Subnet contains the IP addresses of the VMs.

VPN Gateway supports Availability Zones.

VPN Gateway SKU

learn.microsoft.com/en-us/azure/vpn-gateway/about-gateway-skus

VPN Gateway Generation	SKU	S2S/VNet-to-VNet Tunnels	P2S SSTP Connections	P2S IKEv2/OpenVPN Connections	Aggregate Throughput Benchmark	BGP	Zone-redundant	Supported Number of VMs in the Virtual Network
Generation1	**Basic**	Max. 10	Max. 128	Not Support	100 Mbps	Not Support	No	200
Generation1	**VpnGw1**	Max. 30	Max. 128	Max. 250	650 Mbps	Supported	No	450
Generation1	**VpnGw2**	Max. 30	Max. 128	Max. 500	1 Gbps	Supported	No	1300
Generation1	**VpnGw3**	Max. 30	Max. 128	Max. 1000	1.25 Gbps	Supported	No	4000
Generation1	**VpnGw1AZ**	Max. 30	Max. 128	Max. 250	650 Mbps	Supported	Yes	1000
Generation1	**VpnGw2AZ**	Max. 30	Max. 128	Max. 500	1 Gbps	Supported	Yes	2000
Generation1	**VpnGw3AZ**	Max. 30	Max. 128	Max. 1000	1.25 Gbps	Supported	Yes	5000
Generation2	**VpnGw2**	Max. 30	Max. 128	Max. 500	1.25 Gbps	Supported	No	685
Generation2	**VpnGw3**	Max. 30	Max. 128	Max. 1000	2.5 Gbps	Supported	No	2240
Generation2	**VpnGw4**	Max. 100*	Max. 128	Max. 5000	5 Gbps	Supported	No	5300
Generation2	**VpnGw5**	Max. 100*	Max. 128	Max. 10000	10 Gbps	Supported	No	6700
Generation2	**VpnGw2AZ**	Max. 30	Max. 128	Max. 500	1.25 Gbps	Supported	Yes	2000
Generation2	**VpnGw3AZ**	Max. 30	Max. 128	Max. 1000	2.5 Gbps	Supported	Yes	3300
Generation2	**VpnGw4AZ**	Max. 100*	Max. 128	Max. 5000	5 Gbps	Supported	Yes	4400
Generation2	**VpnGw5AZ**	Max. 100*	Max. 128	Max. 10000	10 Gbps	Supported	Yes	9000

VPN Gateway pricing

VPN Gateway Type	Price	Bandwidth	S2S Tunnels	P2S Tunnels
Basic	**$26.28**/month	100 Mbps	Max 10 1-10: Included	Max 128 1-128: Included
VpnGw1	**$138.70**/month	650 Mbps	Max 30 1-10: Included 11-30: **$0.015**/hour per tunnel	Max 250 1-128: Included 129-250: **$0.01**/hour per connection
VpnGw2	**$357.70**/month	1 Gbps	Max 30 1-10: Included 11-30: **$0.015**/hour per tunnel	Max 500 1-128: Included 129-500: **$0.01**/hour per connection
VpnGw3	**$912.50**/month	1.25 Gbps	Max 30 1-10: Included 11-30: **$0.015**/hour per tunnel	Max 1,000 1-128: Included 129-1,000: **$0.01**/hour per connection
VpnGw4	**$1,533**/month	5 Gbps	Max 100 1-10: Included 11-100: **$0.015**/hour per tunnel	Max 5,000 1-128: Included 129-5,000: **$0.01**/hour per connection
VpnGw5	**$2,664.50**/month	10 Gbps	Max 100 1-10: Included 11-100: **$0.015**/hour per tunnel	Max 10,000 1-128: Included 129-10,000: **$0.01**/hour per connection

VPN Gateway pricing

Pricing depends on:

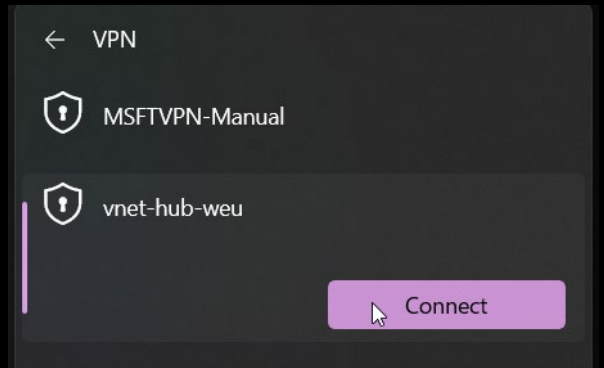
1. SKU
2. Data transfers

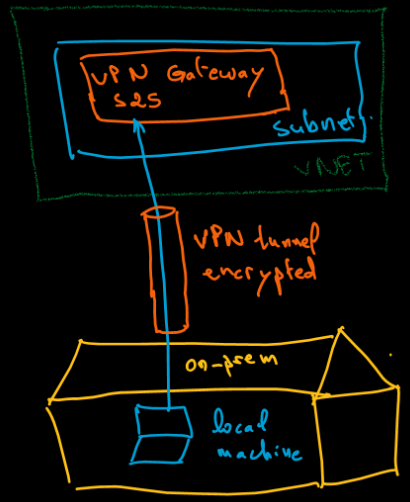
azure.microsoft.com/en-us/pricing/details/vpn-gateway

Type	Example	Price
Inbound Inter-virtual	Data going into Azure data centers between two virtual networks	Free
Outbound Inter-virtual	Data going out of Azure data centers between two virtual networks	From Zone 1*— $0.035 per GB From Zone 2*— $0.09 per GB From Zone 3*— $0.16 per GB
Outbound P2S (Point-to-Site) VPN	Data going out of Azure Virtual Network via P2S VPNs	Data transferred out of Azure Virtual Networks via the P2S VPNs will be charged at standard data transfer rates.

Creating VPN Gateway

1. Create Subnet GatewaySubnet
2. Create VPN Gateway
3. Generate Root certificate for authentication
4. Upload root certificate to VPN Gateway
5. Specify CIDR range for clients
6. Download VPN Client
7. Connect to VPN Gateway





Create virtual network gateway

Subscription *	Microsoft-Azure-T
Resource group	rg-hub (derived from virtual network's resource group)
Instance details	
Name *	vnet-gateway
Region *	West Europe
Gateway type *	VPN / ExpressRoute
SKU *	VpnGw2
Generation	Generation2
Virtual network *	(New) vnet-hub Create virtual network
Public IP address	
Public IP address *	Create new / Use existing
Public IP address name *	pip-vpn-gateway
Public IP address SKU	Standard
Assignment	Dynamic / Static
Enable active-active mode *	Enabled / Disabled
Configure BGP *	Enabled / Disabled

Connecting to VNET resources using VPN Gateway

https://github.com/HoussemDellai/azure-network-course/tree/main/805_vpn_gateway_p2s_access_azure

Connecting to Spoke VNET resources using VPN Gateway

https://github.com/HoussemDellai/azure-network-course/tree/main/805_vpn_gateway_p2s_access_azure